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SECTION 2

LAB 2

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TABLE OF CONTENTS

1.0 INTRODUCTION.....	3
2.0 PROJECT BACKGROUND.....	3
3.0 TARGET USERS.....	4
4.0 MANDATORY MODULES.....	5
5.0 INDUSTRY MODULES.....	6
6.0 MODULE DISTRIBUTION TABLE.....	10
7.0 JUSTIFICATION.....	11
8.0 FINANCIAL REQUIREMENTS AND COST ESTIMATION.....	12
8.1 ONE-OFF COSTS.....	12
8.2 YEARLY COSTS.....	12
9.0 CONCLUSION.....	14

1.0 INTRODUCTION

The stakeholder for this project is Dr. Norhanisah Abdullah, the Principal of Genius Aulad Skudai, who highlighted several challenges in managing teachers efficiently within the school. To address these issues, the principal plans to develop a teacher management system called GeniusAqil OS for Tadika Aqil Miqail.

The proposed system aims to improve the management of teacher-related processes, including maintaining teacher records, assigning tasks, monitoring teacher performance, and handling leave applications. By digitalizing these processes, the system is expected to reduce administrative workload, minimize human errors, improve transparency, and ensure fair task distribution among teachers.

2.0 PROJECT BACKGROUND

The current management of teacher-related processes such as maintaining teacher records, assigning daily tasks, tracking teacher performance, and handling teachers' leave are handled manually by the principal without the support of a dedicated system. This manual approach makes it difficult to organize and retrieve information efficiently, increases the risk of human error, and consumes a significant amount of time.

As the number of teachers and responsibilities grows, it becomes increasingly challenging for the principal to ensure that all records are accurate, tasks are fairly distributed, and teacher activities are properly monitored. For instance, some teachers may unintentionally receive additional tasks compared to others, leading to feelings of unfairness and dissatisfaction. Furthermore, the lack of a centralized system reduces transparency and makes monitoring workloads more complicated. Consequently, the administrative process becomes inefficient, error-prone, and less effective in supporting fair and systematic teacher management.

The goal of this project is to develop a comprehensive Android-based teacher management application called GeniusAqil OS that streamlines and centralizes all teacher-related operations. The system aims to improve efficiency by digitizing teacher records, automating task and duty assignments, tracking leave and training, and monitoring performance. By providing a structured and integrated platform, the application will help school management

save time, enhance decision-making, and ensure better communication and accountability among teachers.

3.0 TARGET USERS

This application is designed for the following target users:

1. Teacher
 - Update personal information and upload documents
 - Apply for leave and track leave balance
 - View and complete assigned duties like cleaning, arrival, and dismissal
 - Submit training records and upload certificates
 - Submit reports such as bullying, safety issues, and system problems
 - Access personal performance report
2. Principal/Owner
 - Manage teacher records and documents
 - Monitor and evaluate teacher performance and KPI
 - Review and approve leave applications
 - Assign tasks and duties
 - Review reports submitted by teachers
 - Access reports and analytics

4.0 MANDATORY MODULES

The application will include the required core module:

1. Login Module

- Allow principals and teachers to securely access the application using email and password
- If the information is correct, the user will be redirected to their respective interface based on their role
- If the password is incorrect, the system will display an error message “Incorrect Password”
- If the user forgets their password, users can use the “Forget Password” feature and will receive an email with the reset password link.

2. Registration Module

- Facilitates new user sign-up with basic data validation, allowing new staff members to be onboarded into the system

3. Manage User Module

- Create new Principal / Owner users
- Teachers are allowed to:
 - View and edit their own profile
 - Apply for leave
 - View duty schedule
 - Submit training records
 - View own performance
 - Submit reports
 - Change password
- Principals / Owners are allowed to:
 - Get full access to all modules
 - Approve teacher registration application
 - Add or delete teachers
 - Approve or reject leave
 - Add, edit, or delete activities for all duties
 - View all reports

- View teacher's training record
- Evaluate teacher's performance
- Change password
- Role-based access control (RBAC)

5.0 INDUSTRY MODULES

There are six modules to the GeniusAqil OS Teacher Management System:

1. Teacher Record

- The Teacher Record module stores teacher information and important documents in one place. The system allows teachers and principals to easily view and manage teacher records.
- Each teacher must upload important documents such as a copy of their Identification Card (MyKad), passport photo, resume or CV, latest academic certificates, medical check-up report, and bank statement header.
- Teachers can view and update their own information when needed. Principals can access and manage all teacher records. The system helps ensure that all documents are complete and up to date. It also allows users to upload new documents and replace old ones when necessary.

2. Leave Management

- Teachers can apply for leave directly through the app by selecting the leave type, filling in the dates and reason, and attaching any required supporting documents such as a medical certificate. The system supports 10 leave types including Annual Leave, Medical Leave, Maternity Leave, Umrah and Haji Leave, and a Birthday Leave that can only be redeemed on or around the teacher's actual birthday.
- Once submitted, the application is sent to the principal or admin for review, who can approve or reject it directly through the system. The teacher receives an automatic notification informing them of the outcome, and if rejected, a reason can be provided.
- Each teacher has a personal leave dashboard that displays their remaining balance for every leave type, which automatically deducts once a leave is

approved. The system prevents teachers from applying for more days than their remaining entitlement, keeping balances accurate at all times.

- All leave records are stored in the system and can be referenced at any time by both the teacher and the admin, giving full visibility into each teacher's leave history and patterns. Automated reminders can also be triggered for things like pending approvals that have not been acted on, ensuring nothing gets overlooked.

3. Task and Duty Manager

- The task and duty manager will automatically assign monthly and daily duties among all teachers fairly. There are four daily duties which are Cleaning Duty, Arrival Duty, Dismissal Duty, Half-full Day Transition Duty and one monthly duty which is Assembly Duty.
- The system takes into account monthly staff changes, such as onboarding new interns. The system will also link duties with teacher attendance so if a teacher is absent, the system automatically reassigns another teacher.
- The system will auto generate a duty table each week and create checklists of tasks for each duties. Teachers can use this checklist as a reference on what needs to be done to complete the task. Teachers must tick on the checklist once a task is completed and upload images as proof to verify that the task is done.
- The app will also send automated reminders to teachers if their task has not been completed. The principal is allowed to change the assignees for all duties and can see the progress of each teacher's tasks progress.

4. Teacher Training Tracker

- The Teacher Training Tracker module helps teachers record and monitor the training programmes they attend. Teachers must enter details such as the training title, category, date, training mode, and reflection.
- They are also required to upload certificates and training photos as proof of attendance.

- The system provides a list of training categories including Teaching Skills, Child Development, Safety and First Aid, Islamic Education, Classroom Management, ICT/Technology, and Others.
- The system tracks each teacher's training progress and shows whether they have met the required training target. Principals can view all training records and monitor staff development.
- The module also generates annual training reports to support teacher evaluations and management reviews.

5. Teacher Performance Tracker

- Assesses teachers against a list of KPIs via a check-list observation system that automatically calculates performance scores, automatically tracks disciplinary issues, and generates annual reports.
- Allow principal to evaluate teachers KPI using ten performance categories such as attendance, classroom management, teaching performance and professionalism to monitor teacher performance
- Allow principal to monitor classroom standards by checking classroom cleanliness, student discipline, learning environment, and compliance with safety rules
- Allow principal to track student development by recording how teachers support weak students, encourage participation, and maintain positive student behaviour
- Allow principal to assess teaching quality by reviewing lesson plans, teaching aids, classroom engagement, and lesson delivery effectiveness
- Allow principal to manage documentation and record keeping by ensuring attendance records, student files, and assessment records are updated and organised properly
- Allow principal to track task and duty responsibilities such as assembly duty, cleaning duty, school events, and dismissal supervision
- Allow principal to measure creativity and self-development by tracking teacher participation in training, teaching innovation, and initiative in classroom improvement activities

- The system will automatically calculate performance scores, track disciplinary issues, and generate teacher performance reports for annual evaluation purposes

6. Teacher Reporting System

- A digital form system where teachers can submit issues in the workplace, complaints and facility damages for management review.
- There are 10 report types that the teachers can submit. This include sexual harassment, bullying, conflict between staff, SOP violation, workload stress, teacher misconduct. Teachers can also submit reports regarding operational issues such as facility maintenance, teaching material shortage, safety hazard, IT or System problem.
- The system implements anonymous reporting so that the teachers can confidentially submit reports regarding sensitive issues such as bullying and harassment between other staff members.
- Teachers can also attach images as evidence to their report. For example, teachers can upload pictures for damaged facilities like a broken fan for maintenance.
- The system includes a status tracking feature for each report to provide visibility on the progress of the report. Each report can have a status of Pending, Under Review, or Resolved.
- Automated push notifications are sent to notify the principal if there are any new reports and to notify teachers to update on the status of the submission.

6.0 MODULE DISTRIBUTION TABLE

Module	Type	Lead Member
Login	Mandatory	Cheryl
Registration	Mandatory	Aflah
Manage User	Mandatory	Jia Lin
Module 1: Teacher Records	Industry	Auni
Module 2: Leave Management	Industry	Aflah
Module 3: Task and Duty Manager	Industry	Najma, Jia Lin
Module 4: Teacher Training Tracker	Industry	Auni, Aflah
Module 5: Teacher Performance Tracker	Industry	Cheryl, Najma
Module 6: Teacher Reporting System	Industry	Najma

7.0 JUSTIFICATION

Our team divided the modules carefully based on each member's skills, interests, and to make sure the workload is fair for everyone. For the mandatory features like login, registration, and user management, we assigned one person to each module. This is because these parts need clear responsibility, so it is easier to manage, fix errors, and avoid confusion during development. For the more complex modules, such as task and duty management, teacher training, and performance tracking, we assigned two members to work together. These modules involve more features and require more time, so having two people allows better teamwork, idea sharing, and problem-solving. It also helps reduce mistakes and ensures the module can be completed more efficiently. In addition, we considered each member's interest and understanding when assigning the industry-related modules. By doing this, members can work on parts they are more familiar with, which can improve the quality of the system.

8.0 FINANCIAL REQUIREMENTS AND COST ESTIMATION

The financial framework for the GeniusAqil OS project is divided into two distinct phases, the one-off development costs required to build and deploy the system, and the recurring annual costs necessary for cloud infrastructure and technical maintenance.

8.1 ONE-OFF COSTS

The proposed initial capital expenditure as shown in Table 8.1 is based on the 2-months (8 weeks) delivery timeline. The project estimated total work hours is 320 hours, with a market rate of RM80 per hour. This budget covers the intensive frontend engineering, NoSQL database structuring, and the complex logic required for automated features in the system and role-based access control. It also takes into account the one-off cost needed for a Google Play Platform Account.

Item Description	Details	Estimated Cost (RM)
System Architecture and UI/UX Design	Labor (60 hrs)	4,800.00
Cross-Platform Frontend Development	Labor (100 hrs)	8,000.00
Backend Engineering	Labor (100 hrs)	8,000.00
QA, Testing and Deployment	Labor (60 hrs)	4,800.00
Google Play Platform Licensing	One-Off Fee	115.00
Total Cost		RM25,715.00

Table 8.1: One-Off Cost Estimation for GeniusAqil OS Development

8.2 YEARLY COSTS

The Table 8.2 outlines the recurring annual costs required to maintain GeniusAqil OS post-deployment. A portion of the budget is dedicated to secure Cloud Storage and database. Additionally, the budget includes the mandatory Apple Developer Program licensing and a Technical Maintenance SLA to provide continuous bug fixes and security patches as Android and iOS platforms evolve.

Item Description	Details	Estimated Cost (RM)
Apple Developer Program Licensing	Annual subscription (\$99 USD)	470.00
Cloud Database & Functions (Firebase)	Authentication, real-time syncing, and push	600.00

	notifications.	
Cloud Storage	Secure storage capacity for high-volume uploads (certificates, daily duty photo proofs).	350.00
Technical Maintenance SLA	Annual Service Level Agreement covering bug fixes, server monitoring, and OS compatibility updates.	2,400.00
Total Cost		RM3,820.00/year

Table 8.2: Yearly Cost Estimation for GeniusAqil OS Infrastructure and Maintenance

9.0 CONCLUSION

The purpose of this report is to propose a feasible Android based teacher management system named GeniusAqil OS, which aims to digitalize and centralize all teacher related administrative processes of Tadika Aqil Miqail, Genius Aulad Skudai. The system has three compulsory modules and six industry modules built in an academic scope where the costs estimated for total development is RM25,715.00 while for the annual maintenance is RM3,820.00, comprising cost of cloud infrastructure, platform licensing, and technical support. The project is currently being scoped on the Skudai branch to test and refine, with scalability being a key design element to enable future expansion via modular architecture, multi-branch user management and centralized data handling. Upon successful implementation, GeniusAqil OS is expected to be able to streamline the administrative tasks, increase transparency, and be a replicable method for teacher management throughout the wider Genius Aulad network.